

Mario Avolio

AI & CLOUD ENGINEER · GENERATIVE & AGENTIC AI

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Summary

AI & Cloud Engineer focused on production-grade generative AI: LLM applications, agentic workflows, and RAG pipelines, served as containerized microservices on the cloud. Research background in computer vision and model optimization, with a Best Paper Award at ICIAP 2025. I build reliable, evaluable systems that move from prototype to production.

Experience

Machine Learning Reply

Milan, Italy

MACHINE LEARNING ENGINEER

Jan. 2026 - Present

- Build agentic workflows, RAG pipelines, and LLM-powered automation for enterprise use cases, from prototyping to deployment.
- Deploy and operate generative-AI services on cloud infrastructure with containerization, CI/CD, evaluation, and observability.

Imaging and Vision Laboratory (IVL), University of Milano-Bicocca

Milan, Italy

RESEARCH FELLOW

Jan. 2025 - Dec. 2025

- Build a large-scale synthetic Side-Scan Sonar dataset of 20,000+ labeled images for underwater mine detection, beyond the scale of existing public datasets.
- Applied super-resolution models (SPAN, DRCT) and trained lightweight YOLOv11n/s detectors, recovering detection performance comparable to high-resolution data in low-visibility conditions; Best Paper Award at ICIAP 2025.

Skills

Generative AI & LLMs

LLM applications, RAG, agentic & multi-agent workflows, prompt & evaluation design, fine-tuning (LoRA/PEFT), vector databases

Frameworks & Tooling

LangChain, OpenAI Agents SDK, Hugging Face, OpenAI API, ChromaDB, Pydantic

Deep Learning & CV

PyTorch, self-supervised learning (DINOv2), super-resolution, model optimization (quantization, pruning, distillation)

Cloud & MLOps

Azure, Docker, FastAPI, CI/CD (GitHub Actions), uv, pytest, model deployment & observability, Linux

Programming

Python, SQL, Bash, Java, C++, JavaScript, R

Languages

Italian (native), English (fluent)

Projects

RAG · Multi-agent · Cloud

AI MICROSERVICES HUB

github.com/MarioAvolio/Portfolio

- A FastAPI gateway fronting independent generative-AI microservices: RAG (LangChain + ChromaDB), multi-agent research (OpenAI Agents SDK), and text intelligence. Each containerized with its own tests and CI; Python, uv, Docker Compose.

Education

University of Milano Bicocca

Milan (MI), Italy

M.S. IN COMPUTER SCIENCE (GRADE: 110/110 CUM LAUDE)

Sept. 2021 - March 2024

- **Thesis:** *Self-Supervised Learning and Model Adaptation for Facial Attribute Classification* (DINOv2, LoRA, PEViT).

University of Calabria

Arcavacata (CS), Italy

B.S. IN COMPUTER SCIENCE (GRADE: 108/110)

Sept. 2018 - Sept. 2021

Research Publications

Large-Scale Synthetic Side-Scan Sonar Dataset Generation and Super-Resolution Enhancement for Underwater Mine Detection [Best Paper Award]

ICIAP 2025, Rome, Italy

FIRST AUTHOR

Sept. 2025

From Vision to Execution: Reasoning in Hybrid Intelligent Robots

KR 2023, Rhodes, Greece

CO-AUTHOR

Sept. 2023